

ONNX Converters

June 17th 26

What's new in 2026?

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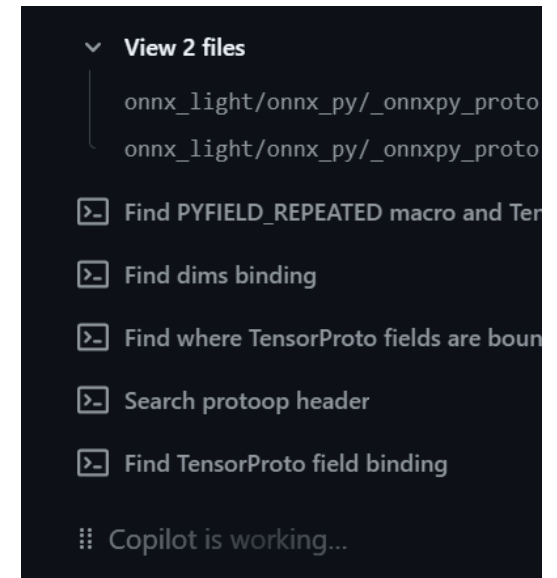
Coding with Agents

The quality of agent writing code reached a gap in 2026 where converting a model into ONNX is just a matter of translation.

It is reliable and it is a lot faster than posting an issue and waiting for it to be resolved.

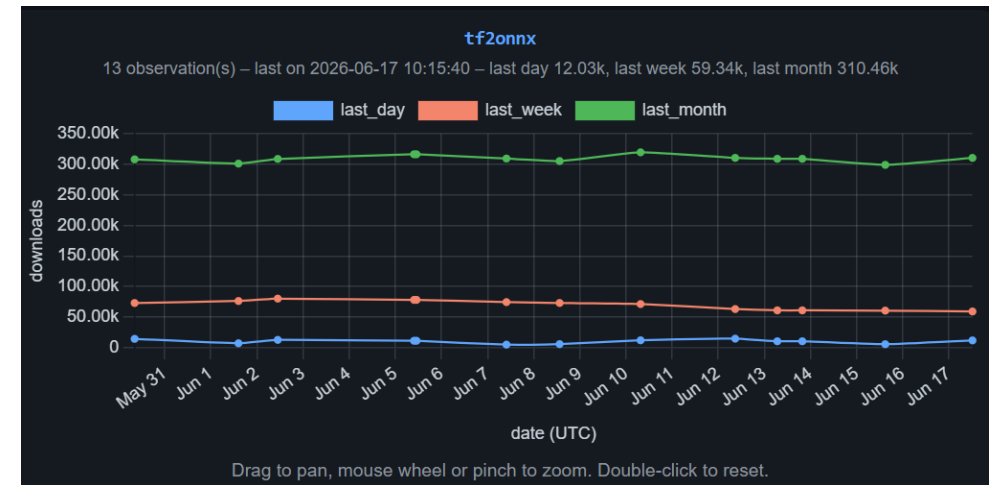
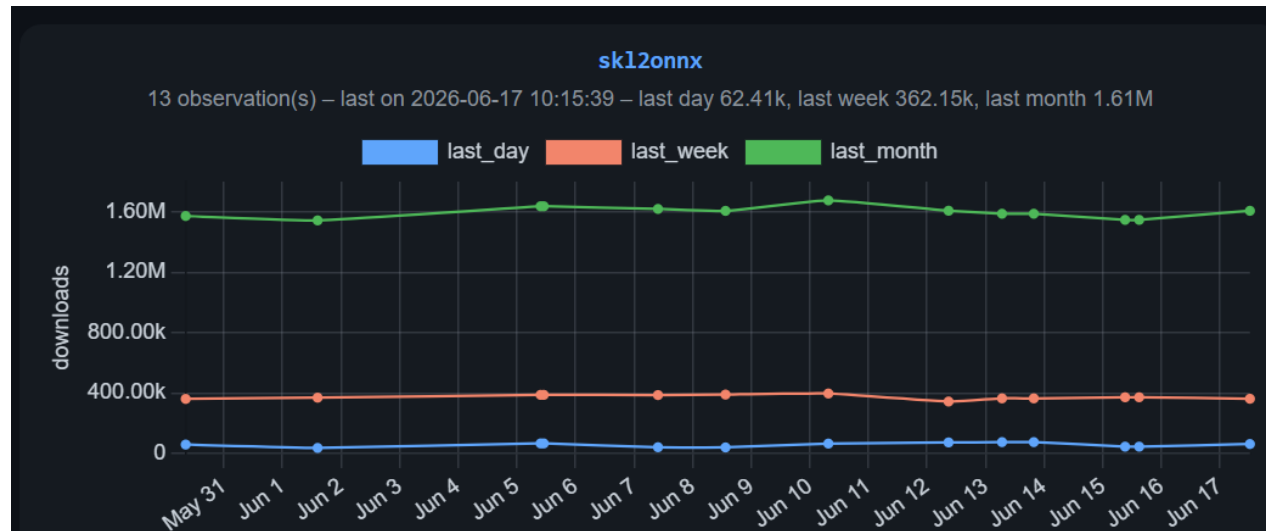
The model is as expected and does not need post-processing.

It works with every machine learning library.



Side effects

If sklearn-onnx, tensorflow are still downloaded on PyPI, we see less and less issues.



Add a new converter in no time

- Create an issue on GitHub, assign it to an Agent.
- Three converters added in a day.



Commits on Feb 24, 2026

Expose <code>parse_sklearn_submodel</code> as public API for meta-estimator parsing (#1233) ...	Verified	8877b73		
Copilot and xadupre authored on Feb 24 · ✓ 24 / 24				
Add ONNX converter for <code>SplineTransformer</code> (#1228) ...	Verified	7772c29		
3 people authored on Feb 24 · ✓ 24 / 24				
Fix <code>float64+OHE</code> mixed <code>ColumnTransformer</code> input type inference regression (#1229) ...	Verified	88f1ad0		
Copilot and xadupre authored on Feb 24 · ✓ 24 / 24				
Fix <code>to_onnx</code> failure on <code>LinearRegressor</code> with multi-column <code>DataFrame</code> input (#1234) ...	Verified	045a456		
Copilot and xadupre authored on Feb 24 · ✓ 24 / 24				
Add ONNX converter for <code>IterativeImputer</code> (#1230) ...	Verified	bc0754a		
3 people authored on Feb 24 · ✓ 24 / 24				
Fix <code>NearestNeighborRadius</code> (#1231) ...	Verified	1c9ef28		
xadupre authored on Feb 24 · ✓ 24 / 24				

Do you still need a framework? Yes

- A model is usually a pipeline of estimators (scikit-learn) or modules (transformers)
- It is safer to export piece by piece.
- Agents cannot do a whole model fast.
- A new model means a couple of new pieces to add.
- Agents need a structure to rely on and a few converters to replicate.
- When it grows, it becomes a framework no matter what you do.

Core Pieces

- You need a GraphBuilder: to add nodes into an ONNX graph.
- A naming logic: how to name the intermediate results
- A logic to register a function to convert a piece.
- A logic to walk through a model, call every converter and connect them.
- A test infrastructure to check discrepancies, speed up and to help the agents automatically check what they do

Optional Pieces

- Optimization
 - The converted ONNX models may not be optimal (remove duplicated transpose, ...).
 - Models are written differently on different providers.
- Shape Inference
 - Allows an optimizer or a runtime to take better decisions.
 - Partially provided by onnx.

Demo: fuse skl2onnx, tf2onnx, onnxmltools

- We don't write code, we review what agents do.
- A common framework speeds up this part.
- No dependency except onnx: you don't want to wait for an issue somewhere to be fixed.
- 95% written with agents.
- Fusing all converters into a single project and share the core and optional pieces.

yet-another-onnx-builder

- [xadupre/yet-another-onnx-builder](#): yet another onnx builder
- Fuse sklearn-onnx, tensorflow-onnx, onnxmltools...
- Extends it to SQL, Pandas, JAX, category_encoders...
- One function to convert all

```
from yobx import to_onnx
expected = model(*args, **kwargs)
onnx_model = to_onnx(model, args, kwargs, dynamic_shapes, **options)
```

- [SQL](#)
- [polars](#)
- [pandas](#)

- [litert](#)
- [jax](#) *in progress*
- [tensorflow](#)
- [torch](#)

- [category_encoders](#)
- [imbalanced-learn](#)
- [lightgbm](#)
- [scikit-learn](#)
- [scikit-survival](#)
- [statsmodels](#)
- [xgboost](#)

New converter

category	estimator	predictable	yobx	converter
category_encoders	BackwardDifferenceEncoder	✓		
category_encoders	BaseNEncoder	✓		
category_encoders	BinaryEncoder	✓	✓	category_encoders_bin
category_encoders	CatBoostEncoder	✓		
category_encoders	CountEncoder	✓		

Copilot AI commented 4 minutes ago • edited ▾

- ☐ Add `basen_encoder.py` converter for `category_encoders.BaseNEncoder`
- ☐ Register the converter in `__init__.py`
- ☐ Add unit tests
- ☐ Validate with black/ruff/mypy and run tests

Issues 6

add converter for category_encoder BaseNEncoder #2290

Open

xadupre opened now

No description provided

Create sub-issue

Assign agent to issue Preview Feedback X

The agent will use the issue description, comments, and any additional prompt you provide to get started. Select a custom agent for specialized tasks.

Optional prompt

Optionally provide additional context for the agent, like constraints, priorities, or areas to avoid

`xadupre/yet-another-onnx-builder` Claude Opus 4.8

Cancel Assign

[Add ONNX converter for category_encoders BaseNEncoder by Copilot · Pull Request #2291 · xadupre/yet-another-onnx-builder](#)

```
from category_encoders import BaseNEncoder
from yobx.sklearn import to_onnx

enc = BaseNEncoder(cols=["cat1", "cat2"], base=3).fit(X_df, y)
onx = to_onnx(enc, (X_df.values.astype("float32"),))
```

Conclusion

Do you still code?